Virtual Labs
An MoE Govt of India Initiative

Data transformation within the MD5 hashing algorithm

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Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Padding:Padding is adding extra bits to a message so that its length is 448 bits short of a multiple of 512. This prepares the message for processing in 512-bit blocks.

- PADDING
- INITIALIZATION
- ADD FUNCTION
- ADD BUFFER A
- ADD SUB-BLOCKS
- ADD CONSTANTS
- CIRCULAR SHIFT
- BUFFER UPDATE
- STORE IN BUFFER B
- BUFFER SHIFT

Enter Your Message

Your Text

MirzaHussain

Generate

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Message Padding

Original Message Bit	Padding Bits	Length of Input
010011.....000000	10000.....00000	000000.....100000

← 512 Bits Block →

512 Bits

Number of Characters: 12
Number of Input Text Bits: 96
Number of Padding Bits: 352

The input message has been successfully padded according to the MD5 algorithm. It is now prepared for chunk processing, where it will undergo transformations to generate the final hash.

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Initialization: The MD5 algorithm initializes four 32-bit buffer variables (A, B, C, and D) with specific constant values.

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A B C D

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Add Function: Each round utilizes a different non-linear mathematical function. These functions are designed to be complex and unpredictable, making it difficult to predict the outcome based solely on the input message.

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Add Buffer A: Buffer A, which stores an intermediate hash state, is added to the current state to accumulate changes over multiple rounds, enhancing security.

PADDING

INITIALIZATION

ADD FUNCTION

ADD BUFFER A

ADD SUB-BLOCKS

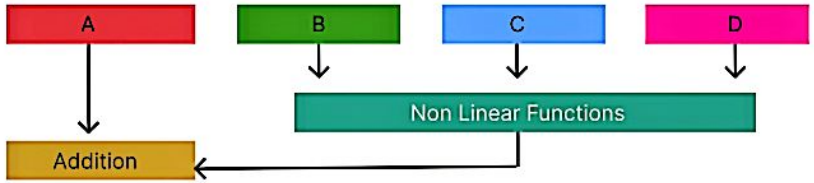
ADD CONSTANTS

CIRCULAR SHIFT

BUFFER UPDATE

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BUFFER SHIFT



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Add Sub-Blocks: The padded message is divided into 512-bit blocks, which are further split into sixteen 32-bit sub-blocks. These sub-blocks are then added to the buffer for processing

PADDING

INITIALIZATION

ADD FUNCTION

ADD BUFFER A

ADD SUB-BLOCKS

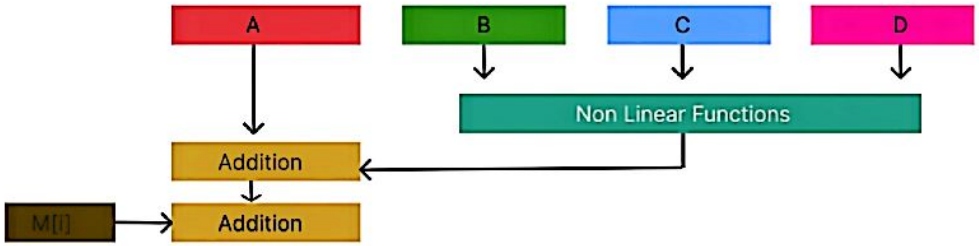
ADD CONSTANTS

CIRCULAR SHIFT

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Add Constants: Predefined constants derived from the sine function are added to introduce randomness and strengthen security.

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ADD FUNCTION

ADD BUFFER A

ADD SUB-BLOCKS

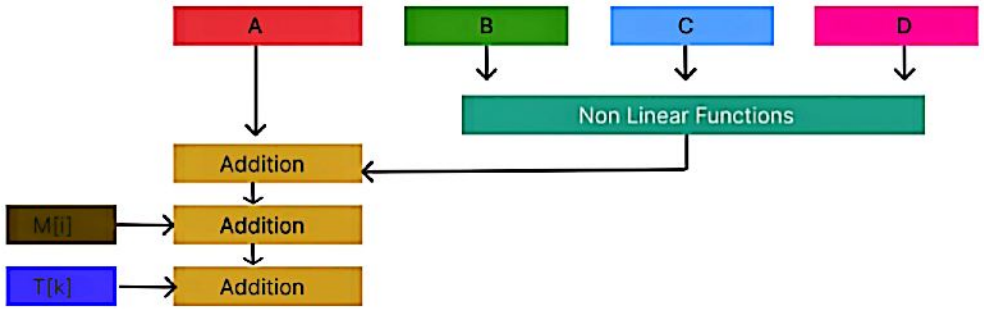
ADD CONSTANTS

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BUFFER SHIFT



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Circular Shift: A circular left shift operation is performed on buffer values to spread changes and improve diffusion

PADDING

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ADD FUNCTION

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ADD SUB-BLOCKS

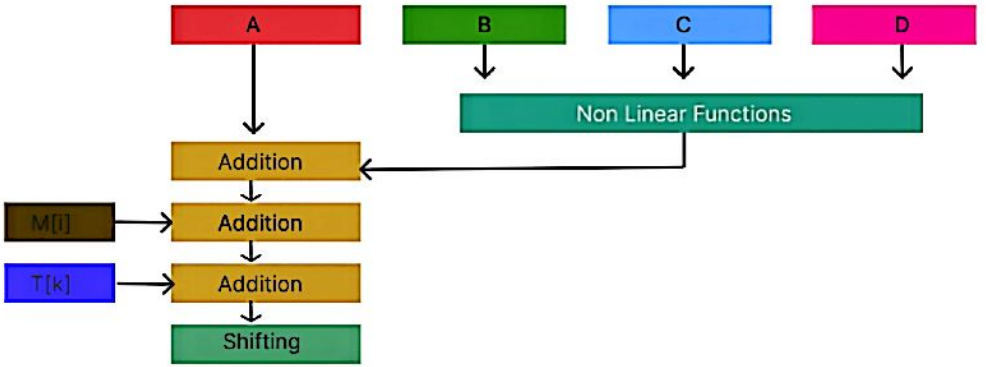
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CIRCULAR SHIFT

BUFFER UPDATE

STORE IN BUFFER B

BUFFER SHIFT



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Buffer Update: The buffer values (A, B, C, and D) are updated based on the processed sub-block, modifying the hash state.

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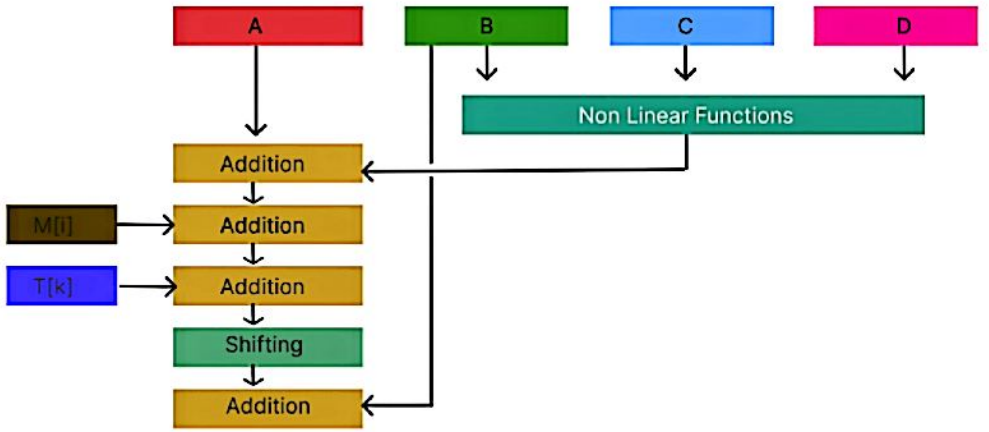
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Store in Buffer B: The updated result is stored in Buffer B for the next step

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ADD BUFFER A

ADD SUB-BLOCKS

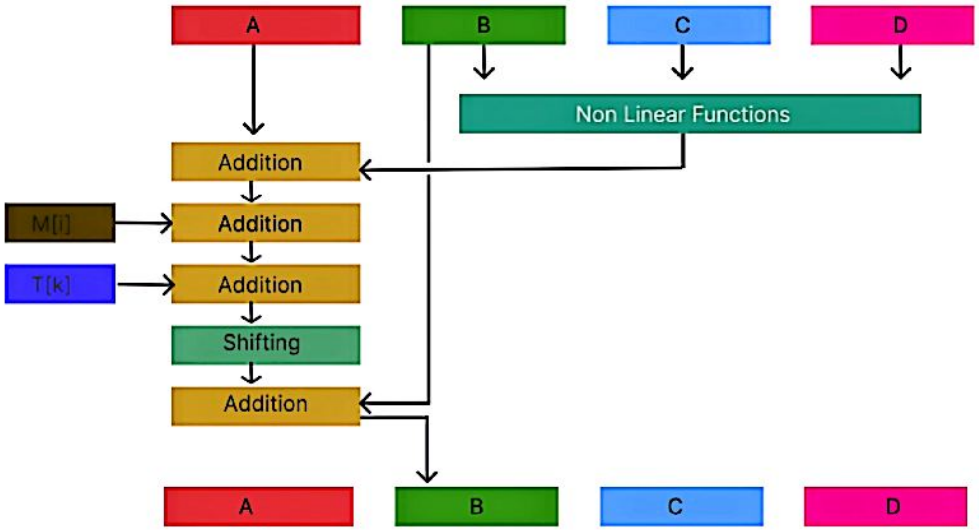
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Buffer Shift: Shift the buffer for the next iteration. This completes one cycle. This process will continue 16 times for each non linear function.

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ADD SUB-BLOCKS

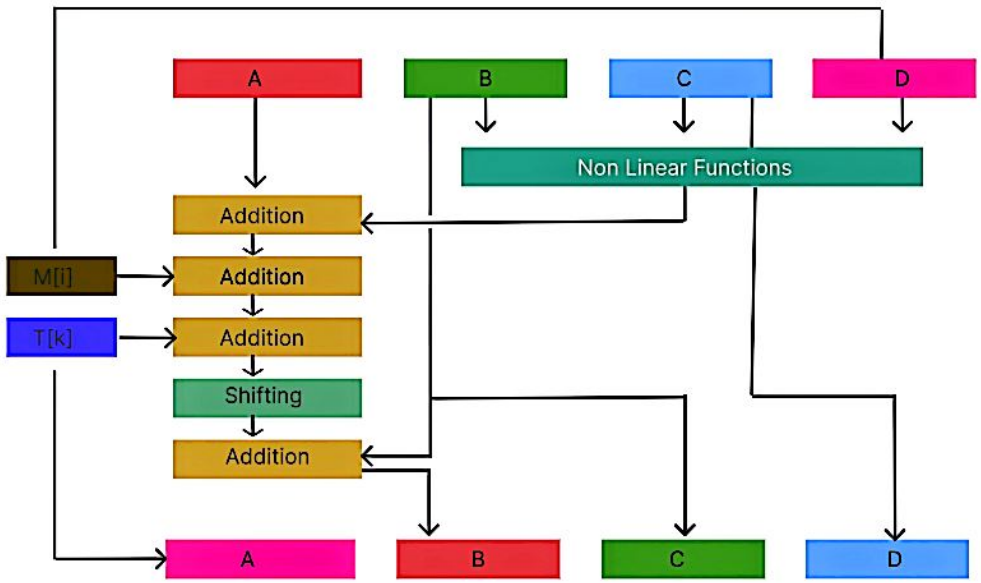
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SHOW MD5 HASH VALUE

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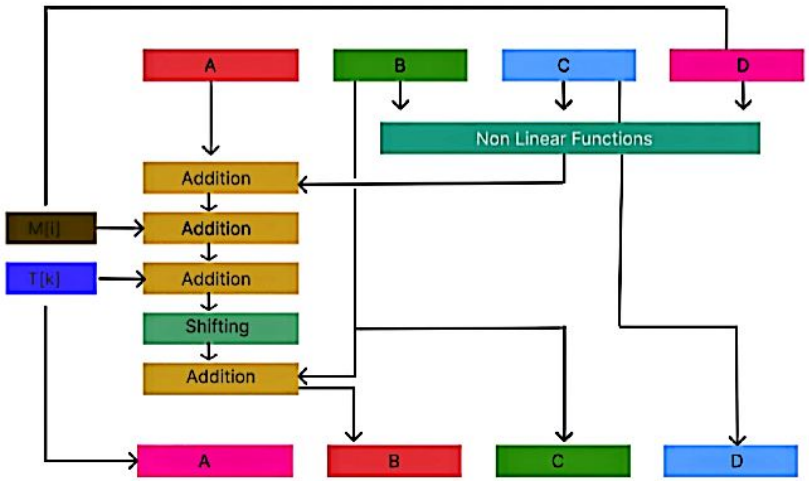
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SHOW MD5 HASH VALUE

836c8e399693899b46e57ff97c982cfe

The MD5 hashing process is complete. Your hashed value is displayed above.